Overview

This document describes the shared components, behavior, and configuration that apply across Khan Academy's Drag and Drop widget family:

- Categorizer
- Composer
- Fill in the Blank
- Sorter

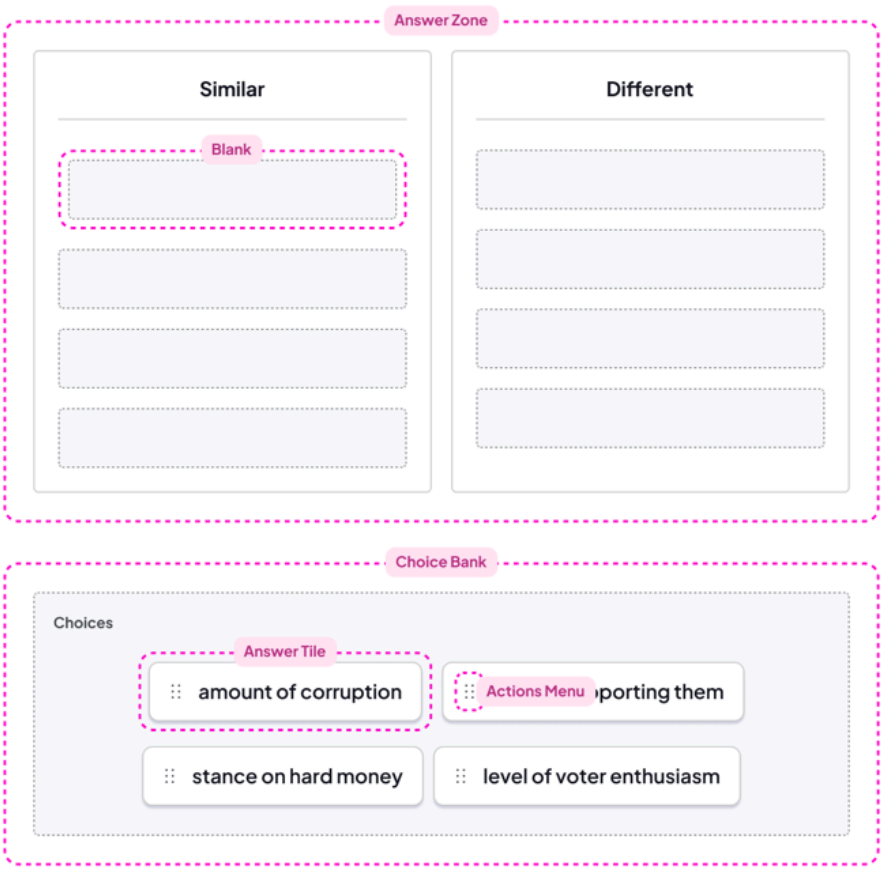
Individual widget documentation builds on this foundation, covering only the behaviors and configurations that differ from or extend what is described here.

Anatomy

Elements

The Drag and Drop widget family shares the following core elements:

Element	Definition
Answer Zone	The active area containing the blank(s) where answer tiles are placed.
Blank	The bay where an answer tile is placed in order to submit an answer for scoring.
Choice Bank	The container of the answer tiles that are moved into blanks for scoring.
Answer Tile	The individual items learners can move into a blank to answer a question.
Actions Menu	The control that allows keyboard / screen reader users to move an answer tile.



Information Hierarchy

Drag and Drop widgets share a layout hierarchy that scaffolds the information so that the learner has appropriate context as they explore the content. The information is presented in the following consistent order: Exercise Stimulus, Answer Zone, and Choice Bank.



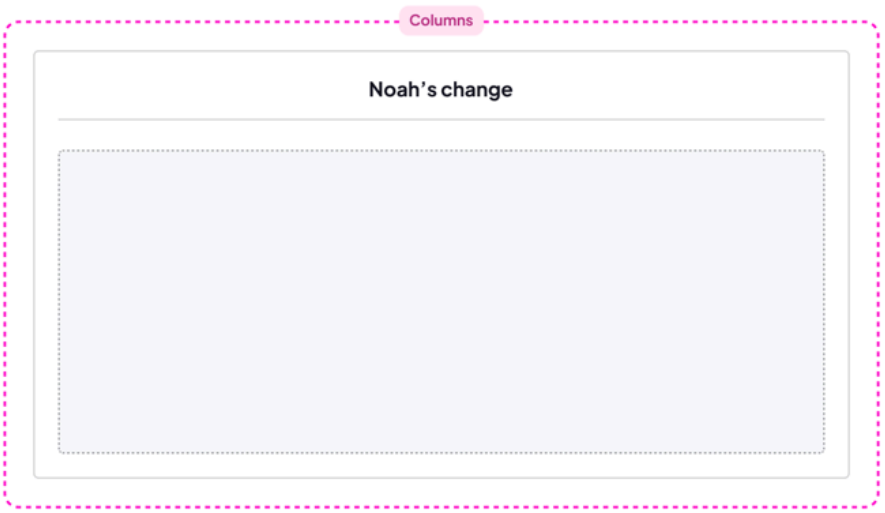
This order is designed for maximum comprehension, presenting the stimulus and context, the question and requested answers, then the choices they can use to satisfy the exercise requirements.

Behavior

Answer Zone

Answer zones vary in format across the widget family. The following formats are used across widgets:

Columns



The Categorizer, Composer, and Sorter widgets have column-based answer zones. The number of columns varies based on the widget.

- Columns display side-by-side, filling the available width of their container. Once a column would drop below ~264px wide, they stack vertically from start to end — sooner when 3 columns are present.
- When columns include multiple blanks, the blanks are contained in an unordered list.
- Labels are required so learners know which tiles to move into each column. Labels fill the container width and reflow onto multiple lines when needed.
- Labels should be specific and unique across columns. When a specific label would compromise the integrity of a single-column exercise, a generic "Answer" label can be used.
- When scored, correctness feedback displays alongside the column header, not inline with the answer tiles.

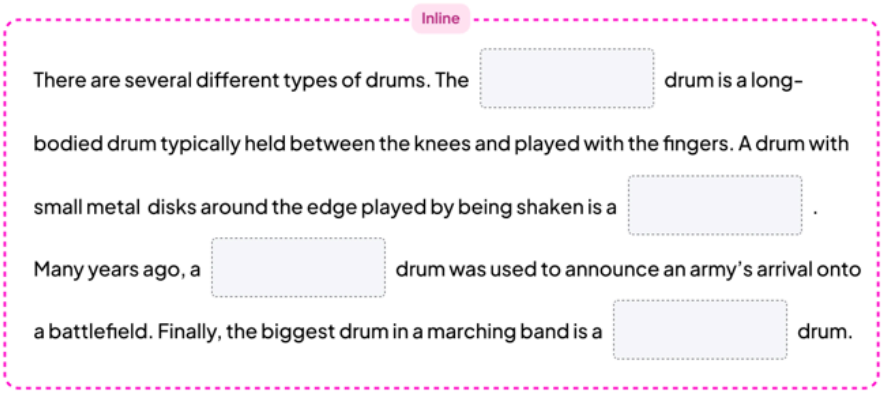
Legend

The Sorter widget has an additional format, the legend format, that provides instructional context in the answer zone so that learners understand key attributes expected for or relationships between items in the answer zone so that they can successfully answer the question.



Inline

The Fill in the Blank widget utilizes an inline format, where blanks are placed in between other segments of content in the answer zone.



Choice Bank

The choice bank contains the answer tiles that learners can move into a blank to complete an exercise.

- Answer tiles are contained in an unordered list and display inline, wrapping to multiple lines when needed.
- The choice bank takes up 100% of the available width of its parent container, with height hugging its contents.
- When the container is too narrow to fit more than one tile on a line, tiles stack vertically. This may not apply to all choice banks if tiles are narrow enough to fit two per line at small viewport sizes.
- When tiles of multiple heights are present, tiles of equal height should display on the same line where possible. Ideally the tallest tiles appear at the end of the bank, but the randomization requirement may mean this is not possible.

Answer Tiles

Answer tiles are moved into a blank by clicking or touching and dragging over the target blank.

Content types

Answer tiles support Text, TeX, Images, and Empty content types. All tiles in a set should share the same content type to ensure a consistent experience.

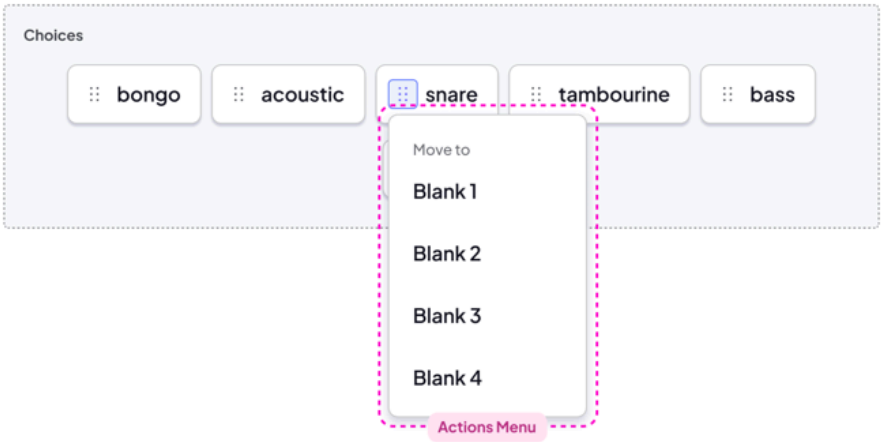
- Text, TeX, and Image types should not be mixed with each other. Empty tiles are allowed alongside other content types.
- Empty tiles have no content and are labeled "(empty)" for screen readers. Design exercises to avoid requiring empty answers where possible. When unavoidable (e.g., balancing a chemical equation), empty tiles are acceptable.
- When an answer tile is multi-use, a screen reader-only label communicates how many uses remain, decrementing until the tile is removed from the bank. Example: "Penny, 5 remaining."

Sizing

Answer tile dimensions adapt to their content and container. Content has a maximum width of 200px to keep tiles readable and consistently sized across a set.

- Tiles hug the height of their content. Width can be set to hug content or fill the parent container, depending on the widget type.
- Text and TeX content reflows to multiple lines when it exceeds the maximum width. Images must not exceed the maximum content width. Content cannot scroll within a tile.
- When tiles stack vertically at small viewports, they fill the available width of their parent container. Text content wraps at the maximum width. TeX wrapping may not be feasible — a character limit may be needed to prevent content from overflowing at small sizes.

Actions Menu



Touch or pointer-based drag and drop interactions are not available to keyboard-only or screen reader users. An Actions Menu button is included on each tile so that these users can

easily and successfully interact with drag and drop items. This menu is also available to touch and pointer users.

Visibility

The actions menu icon button is visible in the answer tile's default state when it is resting in the choice bank and remains so when placed in a blank, with exceptions for the Fill in the Blank and Sentence Variant of the Sorter widget. When an answer tile is used to complete a string (a sentence, paragraph, equation, or formula), the actions menu button disappears when the answer tile is placed in an inline blank so that the string is easier to read without the icon creating a point of interest to disrupt eye movement. Hiding this control from view when a tile is embedded within other content allows maximum readability. The actions menu button re-appears in the tile on hover or focus.

When an answer tile is in the choice bank and the actions menu is activated, it should display above the tile to give the menu the most visible display area. When an answer tile is placed in a blank, the menu displays underneath the tile.

Menu Button Labelling

The actions menu is the first focusable element in an answer tile, which means it should also contain instructive context for screen reader users so that they do not have to traverse the rest of the tile to know what this menu does and what it controls. The actions menu is an icon button that uses aria-labelledby and aria-describedby to associate appropriate context with the control. The button is labeled by the tile value, and the describedby attribute should be associated with screen reader-only content that includes the count remaining for use for multi-use tiles, followed by the label of the button itself "actions menu". When the tile is single-use, no count is necessary.

Single-use example: Bongo. Actions Menu. Multi-use example: Penny. 5 remaining. Actions Menu.

Menu Actions

All answer tiles in the choices bank can be moved to any blank in the answer zone. When an answer tile is placed in a blank, its actions are adjusted to reflect what's possible: moving it to any blank other than its current placement, or clearing it from the placed blank and returning it to the choice bank.

The available actions in the menu change to include the names of the other areas the tile can be placed and a clear action to remove it from a blank and return it to the choices bank. A tile placed in a blank cannot be moved to that blank, so we do not expose that action. This clear action's visible text is "clear", but for screen readers, we append it to include the blank number or column label so that it is clear to which blank this action is scoped ("Clear Blank 1" or "Clear from [Column Label]").

The same pattern applies to move actions: prepend "move to" before the blank information so that the action makes sense to those listening to the menu.

When the widgets have answer columns, the action labels in the actions menu should reference the column labels ("Move to [Column Label]"). When the widgets have individual blanks, the action labels in the actions menu should reference the blank label ("Move to [Blank 1]"). Any custom actions menu needs will be detailed in the individual widget documentation.

Interactions and States

Placing Answer Tiles in Blanks

- The cursor should change to a grab hand when a pointer device is actively dragging an answer tile.
- When an answer tile is moved to a blank that already contains a tile, the tiles are swapped: the moved tile is placed in the target blank and the previously placed tile is returned to the choice bank.
- When an answer tile is placed in a blank via the actions menu, focus should return to the action menu button of the tile in the first position in the choice bank.
- When a tile is moved into a blank via keyboard or screen reader, the Wonder Blocks announcer should announce the name of the moved tile and the location it was moved to — "[Tile] moved to Blank 1" or "[Tile] moved to [Column Label]".
- When a tile is placed in a blank with an existing tile placement and the tiles are swapped, the announcement should also communicate this feedback: "[Tile] moved to Blank 1, [Tile] returned to Choices."

Aborting a Move / Drag

- When a tile is dragged but not placed in an active blank or the choices bank, that tile returns to its source location and the returned tile retains focus.
- When the escape key is pressed during a drag, the drag action is cancelled.
- If a movement initiated by keyboard is aborted, the answer tile action menu should retain focus.

Scoring the Answer

- When a widget is scored correct, any extra blanks in a column answer zone for the Categorizer widget without answers should be removed so that the height of columns is reduced.
- Correct answer tiles are changed to their correct state. The actions menu is not present.
- Any unused tiles in the choice bank should be set to their disabled state, which removes their drag function and actions menu operability.

- If the choice bank is empty when scored correct, the height of the bank should be reduced to a single line so that it takes up less height in a view.
- The choice bank style changes to remove the dashed line border and change the choices label text color so that it is more subdued, placing the greatest visual weight on the correctness feedback in the answer zone.

Configuration

Content Types

Drag and Drop widgets allow flexible content types for both the question with answer zones and choice tiles. Content creators can build these exercises with Text, TeX, Images, and Blank content types. Different content types should not be mixed with one another, especially in Answer Tiles. Text and TeX content types can be mixed within answer zones or answer tiles when warranted, but images should not be mixed with non-image content types.

Images

Images should be used sparingly for Drag and Drop exercises. They are important for assessing specific types of visual skills (such as pattern recognition). Images of text **must not** be used, the Text and TeX content types should be used instead. Images of currency, for example, are an acceptable use for the Image content type because the visual representation is an important correlation for real-world concepts and understanding.

The image content type prefers square images. Content creators can select an image height from one of seven pre-determined values. These presets give the system the appropriate constraints to manage responsive behavior and scaling without complexity. Non-square images are allowed only for landscape images, where the width exceeds the fixed height. The width of these images scales proportionately based on the chosen fixed height. Image widths must never exceed the tile width, they must fully display in the available tile width without any scrolling.

Image height scale

- 24
- 36
- 48
- 60
- 72
- 84
- 96

Blanks

When multiple blanks are present, their screen reader label is appended with their numbering order based on their presentation order within their containing content.

Sizing blanks is tricky and is largely dependent on the widget type. Blank height and width should always match the sizing of the largest answer tile so that the size of the blank does not hint at the answer. Additional customizations of Blanks, including the responsive behavior, are implemented per-widget.

States and labels

- When a Blank does not have an answer, its state is "not answered"
- When a Blank does contain an answer, the answer tile value is announced as the state attribute

A screen reader user that encounters a blank will hear "Blank 1, not answered" or "Blank 1, [value]"

When an answer tile is dragged over a blank, the blank's design changes state to indicate the blank is active and able to receive the tile placement.

Single vs. multi-use

Content creators can configure how many times answer tiles can be used to satisfy a question, choosing between single-use and multi-use options. A choice bank cannot support mixing single and multi-use tiles, as that creates an inconsistent and ultimately confusing experience for learners. Multi-use tiles are configurable so their use is unlimited or has a limit to the number of times they can be used as answers to a given question. Some widgets, such as the Categorizer widget, restrict the use of multi-use answer tiles because that implementation is at odds with the purpose of the widget.

When multi-use tiles have been used the maximum number of times set by the content creator, the answer tile does not appear in the choices bank. Removing one of the answer tiles from the blank causes the multi-use answer tile to display again in the choices bank.